SILICAFLUX

ADVANCED SiO_2-TO-SEMICONDUCTOR R&D

Technical feasibility, materials platforms, process integration and development roadmap

Prepared for the SilicaFlux technology programme

2 November 2026 | Technical research and strategic development report

CORE THESIS

Build a credible silicon-and-photonics platform from high-purity silica, while treating advanced electrochemical reduction and nanostructured silicon as staged research options rather than assumed commercial processes.

STATUS: CONCEPTUAL TECHNICAL REPORT | NOT A PROCESS DESIGN PACKAGE

Report basis and how to use it

This report expands the SilicaFlux concept into a research portfolio spanning mineral feedstock, silicon extraction and purification, wafer engineering, oxide and photonic materials, adjacent semiconductor families, circular recovery, manufacturing intelligence and chip integration. It is designed to support R&D prioritisation, partner discussions and early feasibility work - not to serve as a construction-ready plant design or a foundry process recipe.

Critical distinction SiO_2 is not itself a semiconductor. It can be a mineral feedstock from which elemental silicon is produced, and it can also be a functional insulator, optical cladding or structural material in a finished device. Silicon, silicon dioxide, silicon nitride and silicon carbide are different materials with different manufacturing routes.

Evidence and maturity language

Label	Meaning in this report
Established industrial route	Widely used commercial manufacturing route with existing suppliers and equipment.
Specialist industrial platform	Commercially available, but requires specialised process control, suppliers or foundry access.
Emerging / scale-up	Published demonstrations or early commercial deployment exist; yield, cost, qualification or scale remains a key question.
Exploratory research	Promising scientific concept, but not a reliable basis for near-term product or plant economics.

Purity labels such as “5N”, “9N” or “11N” are shorthand only. They do not replace an element-by-element impurity specification, sampling method, analytical detection limits, and acceptance criteria agreed with the wafer grower or foundry. The report therefore avoids treating a single nominal purity number as universal semiconductor qualification.

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Executive summary

SilicaFlux has a plausible route into semiconductor technology, but the strongest strategy is not to build every process from the mine to the advanced foundry at once. The company should create a focused materials and chip-design programme, validate a first chip through an established foundry, and use the resulting data to decide which upstream and adjacent technologies justify ownership.

Priority	Programme	Why it matters	Recommended posture
P1	Electronic-grade silicon supply chain	Core material route from quartz to silicon wafers; impurity control determines whether the material is usable.	Partner first; own feedstock characterisation and selective process IP.
P1	First SilicaFlux CMOS ASIC	Proves the processor architecture in real silicon without requiring an owned fab.	Design and tape out via mature-node MPW or shuttle.
P1	SiO_2 interface / dielectric engineering	Directly uses silica-derived silicon and oxide in device structures; measurable through test capacitors and transistors.	Joint development with an academic/foundry process team.
P2	Silicon photonics / SiN photonics	A strong adjacent technology using silicon, SiO_2 and silicon nitride; accessible UK prototyping ecosystem exists.	Prototype through MPW/foundry access.
P2	Wafer recycling and circular materials	Can reduce scrap, recover value and build process data; recovered materials must be requalified.	Start with segregated kerf/offcut recovery study.
P3	Direct electrochemical SiO_2 reduction	Potentially reduces carbon-intensive process dependence, but electronic-grade purity and industrial scale are unproven.	Research track only; no base-case plant economics.
P3	SiC, nanowires, porous Si, Si quantum dots	Potentially valuable specialist products, but they are distinct materials/device businesses.	Feasibility projects after core platform is funded.
P2	AI process control	Cross-cutting capability to improve stability, traceability and yield across all material and wafer stages.	Build data architecture early; start with SPC/DOE and assistive models.

Five strategic conclusions

1. High-purity quartz does not automatically produce electronic-grade silicon. Reduction, chemical conversion, distillation, deposition, crystal growth and wafer preparation each create or reveal separate impurity and defect risks.
2. Do not pay a premium for the purest quartz unless a controlled trial proves it reduces total cost or improves final silicon quality. Quartz for fused-crucible manufacture and quartz used as a silicon furnace feedstock may have different economic specifications.
3. Treat SiO_2 as a second product family, not just a feedstock. Its strongest near-term semiconductor uses are thermal oxide/interface layers, deposited insulation, fused quartz components and photonic cladding - each with its own purity and defect specification.
4. Silicon photonics is the most credible adjacent platform to test early. Foundry-compatible 8-inch Si_3N_4-SiO_2-Si demonstrations and UK multi-project-wafer prototyping provide a concrete route to experiments [R8-R10].
5. The first SilicaFlux chip should be fabricated externally. Owning an advanced foundry before demonstrating product-market fit, process expertise and sustained wafer demand would create disproportionate capital and execution risk.

Recommended target architecture

A staged SilicaFlux platform should consist of: (i) high-purity silica characterisation and process development; (ii) partner-produced electronic-grade silicon and wafers; (iii) an in-house materials and device test laboratory; (iv) a mature-node SilicaFlux ASIC; and (v) photonic and oxide test structures fabricated through specialist foundry partners.

Decision Fund a 12-18 month evidence programme around quartz-to-silicon impurity behaviour, oxide interface quality, and a first foundry tape-out. Keep electrochemical reduction, SiC and nanostructures as gated options, not simultaneous factory commitments.

1. Materials map: what SiO_2 can and cannot become

1.1 The material distinctions

Material	What it is	Role in SilicaFlux	Important caveat
Quartz	Crystalline mineral form of SiO_2.	Natural mineral feedstock; high-purity quartz can also support fused-quartz crucibles.	Natural quartz chemistry, inclusions and grain history vary by deposit.
Industrial silica	Quartz-rich sand, rock or other silica material.	Glass, foundry, chemical and other industrial markets.	Industrial “high silica” is not equivalent to semiconductor-grade material.
Fused silica / fused quartz	Amorphous SiO_2 glass made by melting or synthetic chemical routes.	Crucibles, tubes, windows, optical and process components.	OH, bubbles, inclusions, metallic contamination and thermal properties matter.
Elemental silicon (Si)	A group-IV semiconductor element.	Substrate and active material for conventional CMOS.	Must be refined, crystallised, doped and processed; not equivalent to silica.
Thermal SiO_2	Oxide grown by consuming the surface of a silicon wafer.	High-quality interface oxide, isolation and passivation.	Its value depends on interface and reliability, not raw silica purity alone.
Deposited SiO_2	Oxide film deposited by CVD or other processes.	Interlayer dielectric, passivation and optical cladding.	Film density, hydrogen, stress, porosity and contamination need control.
Si_3N_4	Silicon nitride ceramic/film.	Photonic waveguides, masks, passivation and process layers.	Requires silicon- and nitrogen-bearing precursors; not made by simply heating SiO_2.
SiC	Wide-bandgap compound semiconductor.	Power and high-temperature electronics.	Separate crystal-growth and epitaxy ecosystem; quartz is only one possible silicon source.

1.2 The core chemistry

The idealised overall carbothermic reaction is:

$$SiO_2 + 2\,C \rightarrow Si + 2\,CO$$

Industrial submerged-arc furnace chemistry is more complex than this single equation. Silicon carbide and silicon monoxide can occur as intermediates; actual consumption and emissions depend on charge composition, furnace operation, off-gas handling and silicon recovery. The equation is useful for an ideal stoichiometric mass balance, not a complete furnace design model.

1.3 Why the material branches matter

One mineral resource can support several product paths, but the paths have different specifications and economic logic. High-purity quartz for a fused-silica crucible is judged by trace impurities, inclusions, thermal behaviour and crucible performance. Quartz charged to a reduction furnace is judged by reducibility, size distribution, impurity transfer, furnace behaviour and total cost per accepted kilogram of electronic-grade silicon. A single “highest purity” feedstock strategy may therefore be suboptimal.

2. Core SiO_2-to-silicon route and feedstock strategy

2.1 Geological and feedstock qualification

Start with a representative deposit programme, not a mine commitment. Obtain spatially representative samples from multiple lithologies, benches or vein zones; document sample location, chain of custody and preparation; and distinguish bulk chemistry from inclusions and lattice-bound impurities. The USGS describes high-purity quartz as material with less than 100 ppm total impurities (at least 99.99% SiO_2), while ultra-high-purity quartz may contain 10 ppm or less and serves demanding uses such as fused-quartz crucibles [R1]. These are useful market descriptors, not a universal acceptance specification for silicon reduction or semiconductor wafers.

Workstream	Tests / evidence	Decision output
Mineralogy	Petrography, XRD, automated mineralogy, inclusion mapping.	Identify quartz type, gangue minerals, fluid inclusions and liberation limits.
Bulk chemistry	XRF/ICP-MS/ICP-OES with suitable blanks and certified reference materials.	Element-by-element profile: Al, Fe, Ti, Ca, Mg, Na, K, Li, B, P and other relevant elements.

Workstream	Tests / evidence	Decision output
Trace chemistry	High-sensitivity methods selected for B, P and metals; confirm detection limits.	Determine which impurities are likely to persist through beneficiation and reduction.
Physical properties	Particle size, crushability, decrepitation, thermal behaviour, moisture.	Assess processability, furnace burden and breakage.
Variability	Replicate samples and domain-based statistics.	Determine whether supply can meet a stable product specification.

2.2 Beneficiation options

The beneficiation flowsheet should be designed from mineralogical evidence. Potential unit operations include selective mining, staged crushing and screening, optical sorting, magnetic separation, flotation, thermal treatment and chemical leaching. Not every deposit benefits from every operation. The programme must quantify impurity removal against mass yield: a very pure product with poor recovery may be more expensive than a slightly lower-grade feed that behaves better downstream.

A key risk is lattice-bound contamination. Surface cleaning and mineral separation may remove exposed inclusions and gangue, but cannot necessarily remove elements incorporated into the quartz lattice. Such elements may report into silicon or process residues during reduction. Deposit selection should therefore be based on impurity deportment trials, not only bulk SiO_2 percentage.

2.3 Reduction and process mass balance

Theoretical stoichiometry for 1,000 kg of pure SiO_2 is approximately 467 kg elemental silicon, 400 kg carbon consumed and 932 kg carbon monoxide generated. These figures follow the idealised reaction and exclude excess reductant, furnace losses, slag, silicon carbide intermediates, off-gas treatment and actual recovery. The theoretical silicon yield is not a commercial plant yield.

Basis: 1,000 kg pure SiO_2	Theoretical quantity	Interpretation
Silicon produced	467 kg	Maximum elemental silicon from the silica in the ideal reaction.
Carbon consumed	400 kg	Stoichiometric carbon only; actual burden design differs.
Carbon monoxide formed	932 kg	Ideal reaction gas; real off-gas composition and treatment need engineering.
Silica not converted / process loss	Not specified	Must be measured from actual pilot mass balance.
Accepted downstream silicon	Not specified	Depends on furnace recovery and subsequent impurity removal.

2.4 Reduction R&D priorities

- Characterise the reducing carbon independently for B, P, Fe, Al, Ca and other contaminants; the reductant can introduce impurities even where the quartz is clean.
- Compare quartz sources using identical furnace and charge conditions, with full input-output sampling and impurity deportment.
- Measure silicon recovery, specific electricity, electrode consumption, off-gas composition, slag mass and trace-element partitioning.
- Model the product on cost per kilogram of accepted electronic-grade polysilicon, not cost per tonne of metallurgical silicon.

Strategic caution It may be uneconomic to smelt ultra-high-purity quartz directly. A controlled feedstock trial should determine whether additional upstream purity survives the furnace and reduces downstream purification costs. Do not assume that “more pure at the mine” means “cheaper at the wafer”.

3. Alternative silicon purification and direct reduction

3.1 The conventional baseline

For high-purity polysilicon, the modified Siemens chlorosilane route is a robust industrial benchmark. Metallurgical silicon is converted into volatile silicon-bearing compounds such as trichlorosilane; fractional distillation separates impurities; chemical vapour deposition then deposits high-purity silicon. The modified Siemens and silane-based processes remain the principal reference routes in recent reviews [R2-R3]. Their process economics depend on precursor synthesis, distillation energy, deposition rate, reactor utilisation, by-product recycling and product quality.

The route is not a single purification step. Impurities enter or move during raw silicon production, chlorosilane synthesis, distillation, deposition, handling and sampling. A recent review frames quality control across synthesis, purification and reduction as an integrated impurity-management problem [R3].

3.2 A route portfolio, not a silver bullet

Method	Potential function	Main limitation / question	SilicaFlux position
Slag refining	Remove selected metallic and some non-metallic impurities from MG-Si.	Selectivity varies by impurity; slag chemistry, losses and waste management matter.	Bench-scale screening before chemical purification.
Gas blowing / reactive refining	Target selected volatile or reactive impurity species.	Gas handling, kinetics and impurity selectivity; may not solve B/P together.	Research only if a target impurity is identified.
Vacuum refining	Promote removal of volatile species.	Limited for impurities with low volatility in silicon; energy and yield.	Complementary option, not full EG-Si replacement.
Directional solidification	Partition impurities into rejected solidification fractions.	Segregation coefficients vary; B and P can be difficult.	Useful for selected impurity polishing / solar-grade routes.
Zone refining	Move impurity-rich regions along a solid silicon bar.	Slow, costly and geometry-limited; not obvious bulk commodity solution.	Specialist finishing or research samples.
Electron-beam / plasma refining	High-temperature physical refining and volatilisation.	Equipment capital, energy, throughput and contamination control.	Targeted comparative trials only.
Molten-salt electrolysis / FFC	Directly reduce SiO_2 or refine silicon electrochemically.	Electrolyte/electrode contamination, morphology, oxygen removal and scale.	Long-term R&D; not the base-case supply route.
Molten-oxide electrolysis	Explore direct oxide electrolysis in a high-temperature melt.	Materials compatibility, cell stability, anode and product purity.	Exploratory fundamental programme.

A 2026 review surveys metallurgical, chemical, physical, electrochemical and crystallisation routes and explicitly compares purity, energy and environmental trade-offs [R4]. The main strategic lesson is that purification methods have different selectivity. A hybrid route may be valuable, but only when each added stage has a quantified benefit.

3.3 Electrochemical SiO_2 reduction: the high-risk option

The ideal cathodic expression is $SiO_2 + 4e^- \rightarrow Si + 2O^{2-}$. Real cell design must also account for electrolyte composition, oxygen-ion transport, anode reaction, electrode durability, cell containment, thermal gradients, product separation and contamination from cell materials. A successful laboratory reduction does not prove that the silicon product is sufficiently pure or that the process can deliver stable yield at industrial throughput.

The development programme should have staged gates: (1) demonstrate reproducible silicon formation; (2) close oxygen and silicon mass balances; (3) quantify electrolyte and electrode contamination; (4) demonstrate repeatable product morphology and purity; (5) test a downstream polishing route; and (6) carry out a credible energy and cost comparison against carbothermic plus chlorosilane purification. A project should stop or pivot if impurity removal simply shifts the same burden to another expensive stage.

3.4 Recommended baseline architecture

For a near-term SilicaFlux material supply chain, retain conventional carbothermic MG-Si production and modified Siemens purification as the reference case. Test metallurgical pre-refining only where assays show a clear target. Keep direct electrochemical reduction as a ring-fenced research work package with separate funding gates and no assumed contribution to initial production economics.

4. Crystal growth, wafer engineering and quality control

4.1 Czochralski (CZ) growth

The Czochralski process grows a monocrystalline silicon ingot from a melt using a seed crystal. It is a mainstream route for commercial silicon substrates. The quartz crucible and furnace environment affect oxygen incorporation; the axial and radial distribution of oxygen, carbon, dopant and defects can influence downstream processing and device behaviour. A 2026 review specifically examines oxygen behaviour and control in CZ-grown silicon, reinforcing that crucible/melt interactions are a core materials issue [R11].

4.2 Float-zone (FZ) growth

Float-zone growth avoids a quartz crucible during the molten zone and can produce silicon with much lower oxygen content. It is useful for specialist high-resistivity and low-oxygen applications, but diameter, throughput, cost and defect control differ from CZ. It should be evaluated against a defined device requirement rather than presumed superior for all CMOS.

4.3 Crucible engineering as a SiO_2 product opportunity

High-purity fused-quartz crucibles are strategically important because the crucible is part of the silicon growth environment. SilicaFlux could investigate crucible-grade quartz beneficiation, fused-quartz manufacturing or component recycling. However, crucible manufacture requires control of bubbles, inclusions, OH content, metallic impurities, thermal stability and dissolution behaviour. Crucible qualification must be done with an ingot grower; mineral purity alone is not enough.

4.4 Wafer process and measurement

Process stage	Key controls	Measurements / release evidence
Ingot qualification	Crystal orientation, diameter, resistivity, dopant uniformity, oxygen/carbon, dislocation density.	X-ray orientation, FTIR, resistivity maps, defect inspection, dimensional metrology.
Grinding and slicing	Diameter, TTV, bow/warp, saw damage and contamination.	Thickness and geometry maps; surface damage assessment.
Lapping / etching / polishing	Surface roughness, subsurface damage, metal contamination.	AFM/interferometry, particle counts, haze and surface inspection.
Final cleaning and packaging	Particle control, ionic/metal contamination, handling and traceability.	Surface metal analysis, particle inspection, resistivity and wafer genealogy.

A silicon wafer specification is customer- and process-specific. SilicaFlux should not set a generic “semiconductor wafer purity” number without a target wafer type, diameter, resistivity, thickness, orientation, surface finish and defect requirement.

5. SiO_2 dielectric and silicon-interface research

5.1 The key opportunity

Thermal oxidation forms SiO_2 by consuming silicon at the wafer surface. The value is the quality of the Si/SiO_2 interface and the electrically reliable insulating film, not the purity of bulk quartz used as a feedstock. The established semiconductor process sequence uses oxide for insulation, passivation and isolation [R5].

5.2 Distinguish thermal oxide from deposited oxide

Oxide route	How it is formed	Typical R&D questions
Thermal oxidation	Oxygen or steam reacts with the silicon wafer surface.	Interface-state density, oxide thickness, uniformity, stress, fixed charge and breakdown.
CVD SiO_2	Gas-phase precursors react or decompose to deposit an oxide film.	Step coverage, film density, hydrogen content, stress, conformality and contamination.
ALD oxide / high-k stack	Self-limiting surface chemistry deposits ultrathin dielectric layers.	Nucleation, interfacial layer, thickness control, leakage and defect density.

5.3 Scaling and high-k materials

As conventional gate dielectrics become extremely thin, direct tunnelling leakage becomes a scaling constraint. Modern scaled CMOS therefore commonly uses high-k dielectrics such as HfO_2 with a thin interfacial layer and metal gate, rather than relying on SiO_2 alone as the full gate dielectric [R23]. The SiO_2/Si interface remains important, and interface defects can be created or modified during high-k/metal-gate processing [R12-R13].

5.4 SilicaFlux oxide research matrix

Test structure	Purpose	Primary measurements
MOS capacitor	Screen oxide/electrode interface and dielectric quality.	C-V, I-V, hysteresis, flat-band shift, interface trap density.
Gate oxide test transistor	Measure actual device behaviour.	Threshold voltage, subthreshold slope, mobility, leakage, bias-temperature stability.
Metal-insulator-metal capacitor	Separate film dielectric behaviour from silicon channel effects.	Capacitance, leakage, breakdown distribution, time-dependent dielectric breakdown.
Oxide-on-silicon photonic test coupon	Assess optical loss and film uniformity.	Ellipsometry, refractive index, propagation loss, absorption and stress.

Research hypothesis SilicaFlux can potentially build know-how in high-quality silicon oxidation and oxide stack control, but it cannot assume that natural quartz chemistry determines transistor oxide performance. The controlling variables are surface preparation, oxidation/deposition chemistry, process contamination, interface states and reliability.

6. Silicon-on-Insulator (SOI) platforms

6.1 Structure and function

SOI wafers place a thin single-crystal silicon device layer above a buried oxide (BOX), supported by a silicon handle wafer. The BOX reduces parasitic capacitance and improves electrical isolation; SOI can also provide useful etch-stop and radiation-tolerance properties in selected applications [R6].

Conceptual cross-section:

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ACTIVE SINGLE-CRYSTAL SILICON  -  transistor / photonic device layer
BURIED SiO2 OXIDE  -  electrical isolation
SILICON HANDLE WAFER  -  mechanical support
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6.2 Manufacturing routes

Route	Principle	Trade-offs
Wafer bonding / layer transfer	Bond a thin device-quality silicon layer to an oxidised handle wafer; split or thin to target layer.	Flexible device-layer and BOX thickness; bonding quality, defects, cost and process complexity.
Smart Cut-type transfer	Hydrogen/helium implantation defines a cleavage plane; bonding and splitting transfer a thin silicon layer.	Commercially established family of processes; implantation damage and surface finishing must be managed.
SIMOX	High-dose oxygen implantation followed by annealing forms a buried oxide.	Direct oxide formation within silicon; high implant dose/anneal and defect trade-offs.

6.3 SilicaFlux application cases

- Low-power or RF circuitry where reduced parasitic capacitance and isolation are beneficial.
- Radiation-tolerant control electronics for aerospace, scientific or harsh-environment applications.
- Photonic platforms based on SOI, where a silicon device layer is used with a buried oxide optical cladding.

6.4 Qualification

SOI qualification should include device-layer thickness and uniformity, BOX thickness and continuity, interface quality, crystal defects, wafer bow/warp, surface roughness and electrical isolation. The 2023 SOI review provides a useful comparison of SIMOX and layer-transfer routes and discusses wafer characterisation [R6].

7. Silicon photonics and Si_3N_4-SiO_2-Si platforms

7.1 Why photonics belongs in the SilicaFlux portfolio

Silicon photonics uses silicon for optical waveguides and devices, while SiO_2 can act as a lower-index cladding, buried oxide or interlayer dielectric. Silicon nitride (Si_3N_4) supports low-loss waveguides, resonators and nonlinear photonic functions. This materials stack connects naturally to SilicaFlux expertise in silicon, silica and precision thin-film processing.

7.2 What the platform can make

Device / building block	Function	Potential market
Waveguides	Route optical signals on-chip.	Optical I/O, telecom, datacentre links, sensing.
Grating / edge couplers	Couple light into and out of chips.	Fibre-to-chip interfaces.
Ring resonators	Filter, sense or process optical signals.	Wavelength routing, sensing, frequency references.
MZIs and splitters	Interfere, split and combine optical signals.	Modulators, switches, coherent circuits.
SiN nonlinear resonators	Generate combs and support nonlinear optics.	Frequency conversion, metrology, quantum photonics.
Photodetector integration	Convert optical signals to electrical signals.	Optical receivers; may require germanium or heterogeneous materials.

7.3 Evidence of foundry-compatible manufacturing

A 2024 Applied Physics Letters paper reported an 8-inch foundry-fabricated 300-nm Si_3N_4 platform with very low propagation loss and a process design kit [R8]. A 2025 Optics Letters paper reported crack-free 8-inch Si_3N_4-SiO_2-Si wafers fabricated using industrially compatible processes and standard UV stepper lithography, with very low-loss photonic circuits [R9]. These are research demonstrations, not proof that SilicaFlux can immediately reproduce the process, but they show that oxide/nitride/silicon wafer stacks are a credible manufacturing platform.

7.4 Important physical limit

Silicon is an indirect-bandgap material and is not a convenient efficient light emitter in the way direct-bandgap III-V materials are. Practical photonic systems often use external lasers or heterogeneous/hybrid integration for light generation, while silicon and silicon nitride provide routing, modulation, filtering and nonlinear functions. This means SilicaFlux should plan for partner-supplied or integrated III-V sources rather than assuming a pure silicon/silica wafer can do everything.

7.5 Prototype pathway

The most efficient starting point is a multi-project-wafer run through an established photonics platform. The University of Southampton CORNERSTONE service lists SOI platforms at 220, 340 and 500 nm, silicon nitride, suspended silicon and germanium-on-silicon, with DUV-compatible prototyping and MPW access [R10]. This gives SilicaFlux a practical UK route to test optical building blocks without owning a photonics fab. The UK Semiconductor Sector Study 2026 also identifies Southampton and Glasgow as important centres for silicon photonics and foundry prototyping [R22].

8. Silicon carbide (SiC) as an adjacent product family

8.1 Why SiC is different

Silicon carbide is a wide-bandgap semiconductor used in high-voltage, high-temperature and high-efficiency power electronics. It is not a drop-in variation of ordinary silicon CMOS. SiC substrates, crystal growth, epitaxy, defects, wafer finishing and device processes form a distinct technical ecosystem.

8.2 Connection to silica

Silica can be a silicon source for carbothermal SiC production. The idealised reaction is:

$$SiO_2 + 3\,C \rightarrow SiC + 2\,CO$$

This is a useful overall chemistry description, but semiconductor-grade SiC devices require high-purity, single-crystal material. Industrial powder production and growth of low-defect 4H-SiC boules are different operations. Physical-vapour-

transport (PVT) crystal growth and CVD epitaxy are important parts of the 4H-SiC wafer ecosystem. Recent research continues to focus on defect generation and reduction during SiC crystal and epitaxial growth [R14-R15].

8.3 SilicaFlux opportunity

- Near-term: evaluate recovered SiC powder or specialist ceramic/power-material partnerships, not an in-house single-crystal boule plant.
- Medium-term: study SiC substrate and epitaxial wafer supply, defect maps and power-device foundry partnerships.
- Long-term: consider dedicated SiC devices only where customer demand, process know-how and wafer supply are secured.

9. Silicon nanowires, porous silicon and quantum dots

9.1 Silicon nanowires

Silicon nanowires are high-aspect-ratio silicon structures that can provide large surface area and geometry-dependent electrical or optical behaviour. Candidate applications include chemical and biological sensing, photodetection, field-effect sensors and research devices. The main commercial barriers include reproducible morphology, surface passivation, contact resistance, contamination, device-to-device variation and compatibility with CMOS manufacturing.

9.2 Porous silicon

Porous silicon is a silicon network containing engineered pores. Its high surface area can support sensing, electrochemistry and experimental energy-storage electrodes. A 2025 review examines porous silicon and silicon nanowires as candidates for on-chip supercapacitor electrodes, while also noting that planar microcapacitor fabrication and integration remain underdeveloped [R16]. This is therefore an exploratory integration opportunity, not a near-term substitute for conventional on-chip capacitors or external energy storage.

9.3 Silicon quantum dots

Silicon quantum dots are nanoscale silicon domains with size-dependent optical and electronic properties. Research explores ion detection, photocatalysis, photovoltaics, imaging and other applications. A 2025 review identifies promising application areas while highlighting challenges in scale-up, quantum yield and stability [R17].

9.4 Recommended boundary

Keep nanowires, porous silicon and quantum dots in a small, partner-led nanomaterials programme. Define one application and performance requirement before optimising a synthesis method. Avoid a broad “nanotechnology” programme without an identified device, customer or measurable function.

10. Circular materials and recovery systems

10.1 Why circularity matters

Silicon and semiconductor manufacturing generate kerf, offcuts, rejected wafers, crucible scrap, graphite residues, packaging waste and process-chemical residues. Each stream has a different contamination history and recovery route. The business case is not simply that all waste can be returned to electronic-grade silicon; it cannot. The objective is to recover material at the highest technically and economically qualified use.

10.2 Suggested recovery hierarchy

Stream	Potential route	Critical qualification issue
Clean silicon offcuts / rejected wafers	Sort and reclaim for suitable silicon remelting or secondary silicon products.	Metal contamination, coatings, dopants, surface treatments and traceability.
Saw kerf / silicon fines	Recover and purify through suitable chemical or metallurgical routes.	Oxide content, slurry residues, abrasive and metallic contamination.
Quartz crucible scrap	Segregate, assay and assess for silica recovery or non-electronic silica products.	Crucible-use contamination, devitrification and glass chemistry.
Graphite components	Recover/reprocess where grade and contamination allow.	Metal/chemical contamination and oxidation history.

Stream	Potential route	Critical qualification issue
PV module silicon	Dedicated panel recycling and material separation.	Encapsulants, metals, dopants, coatings and mixed provenance.
Chemical process streams	Recover solvents/reagents where validated and safe.	Purity, by-products, cross-contamination and regulatory controls.

A 2024 review of silicon photovoltaic module recycling summarises recovery challenges, material separation and the need for markets for recovered products [R18]. SilicaFlux should not mix solar-grade, electronic-grade, photonic and crucible materials without strict segregation and requalification.

10.3 Circular pilot

1. Create a material-stream register with source, process history, mass, chemistry and contamination class.
2. Choose one low-risk stream first - for example clean silicon offcuts or qualified quartz scrap - and run a laboratory recovery trial.
3. Compare recovery yield, reagent/electricity demand, emissions, waste and value of the recovered grade.
4. Validate the product with a buyer before installing recovery equipment at scale.

11. AI, digital twins and manufacturing optimisation

11.1 The data advantage

SilicaFlux can create a proprietary dataset linking mineralogy and process conditions to final silicon and device performance. Each batch should carry a digital material passport: source location, sample preparation, assay, processing history, furnace inputs, impurity deportment, purification lot, crystal growth conditions, wafer maps, electrical tests and downstream chip yield.

11.2 Where models could create value

Process	Useful modelling tasks	Controls and cautions
Quartz beneficiation	Predict impurity removal and recovery from mineralogy and process conditions.	Models must be calibrated to each deposit and sampling domain.
Silicon reduction	Predict energy, yield, off-gas and impurity partitioning.	Maintain mass-balance closure and retain operator safety interlocks.
Polysilicon purification	Model distillation/deposition performance and trace impurity trends.	Data confidentiality and lab detection limits must be managed.
Crystal growth	Predict oxygen, thermal fields, defect risk and ingot uniformity.	Physics-informed models; validate against metrology and actual growth runs.
Wafer manufacturing	Optimise slicing, polishing, geometry and defect classification.	Avoid overfitting to one tool or one supplier.
Foundry yield	Predict wafer/die yield from process-control and electrical test data.	Respect foundry data rights; use secure, access-controlled datasets.

11.3 Practical implementation

Start with statistical process control (SPC), design of experiments (DOE), traceability and robust measurement systems. Add predictive models where there are sufficient high-quality labelled data. A 2025 IEEE Access paper describes a semiconductor process digital-twin framework combining Bayesian optimisation and deep learning [R19]; a 2026 review surveys AI applications across wafer fabrication, crystal growth, ingot slicing and wafer machining [R20]. These support a research direction, but do not mean AI can replace process engineering, qualification or safety systems.

12. Advanced packaging and heterogeneous integration

12.1 Why packaging should be part of architecture

A modern computing product is a system of dies, memory, power delivery, interconnect, substrate, cooling and test. Advanced packaging allows different chips or chiplets to be integrated in one package, sometimes using silicon

interposers, through-silicon vias, bridges, redistribution layers or hybrid bonding. The IEEE Heterogeneous Integration Roadmap treats packaging as an integrated materials, process, power, cooling and test challenge [R21].

12.2 SilicaFlux chiplet concept

Die / component	Function	First-generation status
SF-CTRL	Scalar control / general-purpose processor.	Core ASIC candidate.
SF-VEC	SIMD/vector engine.	May be integrated on the same die initially.
SF-TENSOR	Matrix/tensor acceleration.	Start as a small functional engine; scale after silicon validation.
SRAM / memory interface	Local storage and external memory control.	Foundry SRAM macros and standard interfaces preferred initially.
Optical I/O / photonics	High-bandwidth optical links.	Separate photonic die or co-packaged optical module at later phase.
Power / clock / test	Power distribution, clocking, scan and test.	Must be designed in from the beginning.

12.3 Technology choices

- Monolithic ASIC: best initial route for proving the architecture, if size and yield are manageable.
- 2.5D package: chiplets placed side-by-side on an interposer or bridge; useful for bandwidth and memory integration, but costly and thermally complex.
- 3D stacking / hybrid bonding: high density and short interconnects, but alignment, thermal management, known-good-die yield and test are difficult.
- Co-packaged optics: potential future route for optical I/O; requires co-design of photonics, electronics, package, laser source and thermal stack.

12.4 Design rule

Do not create a chiplet architecture before the monolithic first-generation design has established real workload, memory and power bottlenecks. Packaging should solve measured system constraints, not become an additional complication without a product requirement.

13. Laboratory, metrology and qualification infrastructure

13.1 Core laboratory capability

Capability	Instruments / methods	Use
Trace elemental analysis	ICP-MS, ICP-OES, XRF, GDMS; suitable sample preparation and blanks.	Quartz, reductants, silicon, polysilicon and contamination mapping.
Mineralogy / structure	XRD, optical microscopy, SEM-EDS, automated mineralogy.	Quartz phase, inclusions, gangue and material defects.
Wafer impurities	FTIR for interstitial oxygen/carbon; SIMS for selected depth profiles; GDMS for bulk trace elements.	Crystal and wafer qualification.
Surface / film metrology	Ellipsometry, reflectometry, AFM, profilometry, XPS, TEM.	Film thickness, roughness, composition, interface and microstructure.
Electrical characterisation	Four-point probe, Hall, C-V, I-V, breakdown and reliability test.	Resistivity, carrier type, dielectric quality and device performance.
Optical metrology	Refractive index, propagation-loss measurement, optical spectrum analysis, resonator Q.	Photonic wafer and device qualification.
Defect inspection	Surface particle scan, laser scattering, X-ray topography and defect imaging.	Wafer, crystal and process defect monitoring.

13.2 Measurement-system discipline

Every reported impurity value must state sample preparation, sample mass, method, detection and quantification limits, calibration, blank level, reference material, uncertainty and whether the result is bulk or surface. “Below detection” is not

the same as zero. Results from different laboratories are not directly comparable unless method and detection limits are aligned.

13.3 Product qualification hierarchy

1. Feedstock qualification: geological variability, inclusion profile and impurity chemistry.
2. Process intermediate qualification: MG-Si, purified chlorosilane, polysilicon and crystal ingot.
3. Wafer qualification: geometry, resistivity, oxygen/carbon, defects, surface and particle cleanliness.
4. Device qualification: electrical characteristics, reliability, yield and customer-specific acceptance.
5. System qualification: packaged-chip functionality, thermal cycling, mechanical reliability and application performance.

14. Industrial facilities, utilities, safety and environmental controls

14.1 Do not confuse a silicon refinery with a wafer fab

Silicon extraction and chlorosilane purification are chemical/metallurgical plants. Crystal growth and wafering are specialist materials plants. Semiconductor fabrication is a cleanroom-based precision process facility. They have different equipment, workforce, utilities, environmental permits and safety cases. Combining all stages on one site is possible in principle but creates complex chemical logistics and contamination interfaces.

14.2 Infrastructure systems

System	Why it matters	Design principle
Cleanroom and air handling	Controls particles, molecular contamination, temperature and humidity.	Classify by process area; segregate chemical, wet and critical lithography zones.
Ultra-pure water	Critical for wafer cleaning and rinse operations.	Validated purification, loop materials, monitoring and microbiological control.
Bulk and specialty gases	Provides process gases, purge and carrier gases.	Gas cabinets, leak detection, automatic isolation, exhaust and abatement.
Chemical distribution	Supplies acids, solvents and precursors.	Compatible materials, double containment, leak monitoring and controlled delivery.
Vacuum and exhaust	Supports deposition, etch and process tools.	Tool-specific pumps, exhaust treatment, corrosion management and maintenance controls.
Power and cooling	Maintains tool stability and protects product during disturbances.	Power quality, backup strategy, chilled water and heat rejection.
Wastewater and abatement	Treats process exhaust and liquid waste.	Segregated waste streams, validated treatment and regulatory discharge control.
Metrology and data infrastructure	Maintains measurement and traceability.	Calibration, SPC, recipe control, historian and secure access.

14.3 Safety and process hazard management

Semiconductor and chemical silicon production can involve corrosive acids, toxic and pyrophoric gases, flammable hydrogen, reactive precursors, high-temperature furnaces, vacuum systems and hazardous residues. OSHA identifies substrate manufacture, device fabrication, metallisation, wafer testing, die separation, bonding and packaging as distinct operations, each with process-specific hazards [R5]. Device fabrication guidance describes oxidation, cleaning, photolithography, etching, doping and deposition operations and their hazards [R7].

The plant concept should therefore require a formal process-hazard analysis; gas detection and automatic shutdown; engineered exhaust and abatement; chemical segregation and compatible containment; emergency response; lockout/tagout; maintenance procedures for contaminated equipment; worker exposure assessment; environmental monitoring; and permit-to-operate review. Detailed recipes and equipment operating conditions must be developed by qualified process-safety and facility engineers.

14.4 Environmental accounting

Track electricity per kg of accepted silicon, carbon and reductant consumption, greenhouse-gas intensity, water consumption, chemical make-up, recycle rates, off-gas treatment, hazardous waste, wastewater load and usable product

yield. Compare processes on a common functional unit - preferably cost and environmental burden per kg of material that meets the actual wafer/foundry specification.

15. SilicaFlux R&D work packages and experimental gates

The programme below converts the research portfolio into bounded work packages. Each work package should have an accountable technical lead, a partner or laboratory, a traceable test plan, a cost ceiling and a gate decision. Quantitative pass criteria should be set jointly with the intended wafer supplier or device foundry after initial baseline measurements.

WP	Scope	Core experiments / deliverable	Gate
WP0	Data and specification architecture	Material passport, impurity dictionary, sample chain-of-custody, data schema and analytical methods.	Can every test result be traced to an identified sample and method?
WP1	High-purity silica characterisation	Map deposit variability, mineralogy, inclusion and trace-element profile.	Is there a repeatable feedstock domain with a credible beneficiation route?
WP2	Beneficiation	Compare sorting, magnetic separation, flotation, thermal and chemical steps where justified.	Does impurity removal justify mass loss and processing cost?
WP3	Silicon reduction	Run partner furnace trials on selected quartz/reductant blends; close mass and impurity balances.	Does feedstock quality improve silicon recovery or downstream impurity load?
WP4	Polysilicon purification	Compare baseline chlorosilane route with selected pre-refining/polishing options.	Can the product meet a buyer-defined electronic-grade specification reproducibly?
WP5	Crystal / wafer	Partner CZ/FZ growth, wafering and material characterisation.	Do ingot and wafer maps meet a specified application/process requirement?
WP6	Oxide / interface	Thermal oxide and deposited oxide test structures; MOS capacitors and interface reliability.	Is there a reproducible dielectric/interface advantage?
WP7	SOI	Procure or partner-fabricate SOI samples; compare bonding/SIMOX platform characteristics.	Is there a device or photonic use case that benefits over bulk silicon?
WP8	Silicon photonics	MPW test chip with waveguides, couplers, resonators and optical I/O blocks.	Are optical loss, yield and PDK integration good enough for a target system?
WP9	SiC	Supplier/partner evaluation of substrate, epitaxy and defect maps; powder route only as separate study.	Is a customer-backed power-device opportunity credible?
WP10	Nanostructures	Select one target device - sensor, nanowire or porous-Si microcapacitor - and benchmark.	Is there a measurable device advantage and reproducible manufacturing route?
WP11	Circular materials	Segregate one waste stream; assay, recover and qualify a secondary product.	Does recovery pass product, EHS and economic qualification?
WP12	AI / digital process control	SPC, DOE, predictive quality and digital twin on one chosen process.	Does model improve yield or variability in a controlled out-of-sample trial?
WP13	ASIC and packaging	SilicaFlux RTL, verification, mature-node tape-out, package/test plan.	Does first silicon meet functional and reliability targets?

16. Roadmap, business model, risks and recommendations

16.1 Proposed roadmap

Phase	Indicative period	Main activity	Exit evidence
A - Define and verify	0-6 months	Lock use cases, sample and assay protocols, PDK/foundry route, processor scope and partner map.	Signed material specification plan; verified RTL baseline; first sample dataset.
B - Demonstrate	6-18 months	Quartz and reduction trials; procure qualified polysilicon/wafers; ASIC MPW; oxide and photonics test structures.	Measured yield/purity data; first silicon; interface/photonic test reports.
C - Optimise	18-36 months	Compare purification route, wafer growth partner, recycling pilot and chip architecture revision.	Cost/performance evidence and second-generation tape-out.
D - Selective industrialisation	3-5 years	Scale only proven materials product or specialist process; secure customers and offtake.	Qualified product, independent customer acceptance and bankable process economics.
E - Foundry or advanced platform decision	5+ years	Assess owned mature-node specialty fab, photonics facility or long-term foundry partnership.	Sustained demand, process IP, experienced team and credible financing.

16.2 Business model options

Model	Description	Capital intensity	Best use
Fabless / foundry partner	Design SilicaFlux chip; outsource wafer fabrication and packaging.	Low-to-medium relative to owning a fab.	First ASIC and architecture validation.
Materials developer	Sell qualified quartz, fused silica or specialist oxide materials.	Medium; depends on process and purity targets.	Earlier materials revenue if customer specifications are achieved.
Polysilicon / wafer joint venture	Co-develop supply with established chemical and crystal-growth partners.	High.	Upstream integration after qualification and offtake.
Specialty process line	Own a niche line for oxide, photonic or wafer component products.	Medium-to-high.	When a differentiated product and customer base exist.
Owned semiconductor foundry	Operate wafer fabrication and process IP.	Very high.	Only after sustained demand and proven process/management capability.

16.3 Principal risks

Risk	Why it matters	Mitigation
Feedstock variability	A single assay can hide spatial variation and mineral inclusions.	Domain-based sampling, representative drilling and independent QA/QC.
Wrong purity economics	High-purity quartz premium may not survive the reduction/purification chain.	Run comparative feedstock trials and cost per accepted EG-Si analysis.
B/P contamination	Some impurities are difficult to remove by simple beneficiation or physical refining.	Element-specific deportment study; compare chlorosilane and refining route.
Electrochemical scale-up	Lab product may not meet purity, energy, throughput or cell-life targets.	Ring-fenced R&D with stop/go criteria.
No foundry design compatibility	An RTL design without a matching PDK and signoff flow is not tape-out ready.	Select fab/PDK early; use foundry IP and design-rule checks.
Photonic integration complexity	Laser source, packaging, thermal and fibre coupling can dominate system performance.	Start with passive PIC test chips and external laser coupling.
Over-diversification	Too many research programmes dilute engineering and capital.	Fund P1 first; stage-gate P2/P3.
Fab capex / utilisation	A wafer fab needs customers, process yield and high tool utilisation.	Use external foundry and MPW before considering ownership.
EHS / permitting	Hazardous chemicals and gases affect design, permits, insurance and operations.	Design safety into concept; specialist HAZID/HAZOP and environmental review.

16.4 Final recommendations

1. Make the first deliverable a functioning SilicaFlux ASIC fabricated by an established mature-node foundry. This validates the computing proposition while keeping the project capital-efficient.
2. In parallel, establish a silica and silicon materials laboratory focused on deposit variability, impurity deportment, analytical quality and cost per accepted wafer-grade output.
3. Treat SiO_2 dielectric/interface work and silicon photonics as the two closest technical adjacencies. They use the silica/silicon materials base and can be prototyped using existing foundry services.
4. Build a circular-materials and digital-traceability layer early. Both improve the economics and evidence base across the rest of the programme.
5. Keep direct electrochemical reduction, SiC crystal production and nanostructured silicon behind explicit technical and commercial gates. They should not be used to justify a near-term foundry investment.

Overall conclusion The strongest SilicaFlux strategy is not “make a semiconductor directly from SiO_2”. It is to build a qualified materials chain from silica to silicon, develop useful SiO_2 and Si_3N_4 device layers, prove a proprietary ASIC through a partner foundry, and expand into photonics or specialist semiconductor materials only where measured device performance and customer demand justify it.

References and selected technical resources

The references below support the process descriptions, technology maturity assessments and selected examples in this report. They are a starting bibliography, not a substitute for a process-specific engineering review, foundry PDK documentation or customer material specification.

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Appendix A. Key formulas and definitions

Item	Expression / definition
Carbothermic reduction (idealised)	$SiO_2 + 2C \rightarrow Si + 2CO$
Ideal silicon yield from pure silica	1,000 kg SiO_2 × (28.0855 / 60.0843) ≈ 467 kg Si.

Item	Expression / definition
Silicon carbide formation (idealised)	$SiO_2 + 3C \rightarrow SiC + 2CO$
Thermal oxidation (simplified)	$Si + O_2 \rightarrow SiO_2$
SOI	A thin single-crystal silicon device layer above a buried oxide (BOX) and silicon handle substrate.
EG-Si	Electronic-grade silicon; acceptance is buyer/process-specific and impurity-specific.
PDK	Process Design Kit supplied by the foundry: process rules, device models, libraries and verification data.
MPW	Multi-project wafer service that shares wafer fabrication capacity among multiple designs.
CMP	Chemical-mechanical polishing, used to planarise wafer surfaces and interconnect layers.
PIC	Photonic integrated circuit.

Appendix B. Suggested first 90-day actions

1. Appoint a technical programme lead covering materials, silicon process and device/foundry integration.
2. Choose one target application for the first SilicaFlux ASIC and freeze a realistic mature-node feature set.
3. Identify three candidate silica feedstocks and commission independent mineralogical and trace-element analysis.
4. Agree analytical protocols, detection limits, sample retention and chain-of-custody rules with the laboratory.
5. Select one foundry/MPW route and obtain the relevant PDK access and design requirements.
6. Commission one silicon oxide test-structure run and one photonics MPW feasibility study through a qualified partner.
7. Build an initial material passport and manufacturing data schema before pilot trials begin.
8. Prepare a cost model comparing external wafer supply, in-house upstream processing and partner-led purification.
9. Create a process hazard and environmental screening register before any chemical or high-temperature pilot facility is specified.